

Harrison Qian

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Education

The Nueva School

San Mateo, CA | 2019 - 2027

Coursework: Linear Algebra, Probability Theory, Mathematical Modeling, Multivariable Calculus, Differential Equations, Computational Biology (Neuroscience), Mechanical Engineering, Algebraic Topology, Quantum Information and Computing

Experience

Research — Anesthetic Depth from EEG, with Dr. Elie Adam ([Adam Lab](#))

Jul 2025 - Present

Building a classifier predicting anesthetic depth from EEG at under 400 parameters, 5000x smaller than comparable models.

Research Intern — [Orbit Engineering](#)

Boston, MA | Jul - Aug 2025

Collected data (EEG, pupillometry, heart rate, fNIRS, PPG, EDA) to characterize Caloric Vestibular Stimulation and Galvanic Vestibular Stimulation. Designed vestibular stimulation waveform to take someone from full alertness to drowsiness in 3 minutes.

Counselor — [Math in the Mountains](#)

San Mateo, CA | Jun - Jul 2025

Worked with 60 mathematically gifted middle schoolers, TA'd number theory and discrete math, led activities like chess.

Projects

Electrical Engineering Explorations | [Timelapse](#)

2025 - 2026

Databox: Designed and soldered a PCB. ESP32 streams potentiometer, microphone, and camera data to Google Sheets over wifi.

Mug Warmer: Built an RC-timed auto-shutoff circuit. 6 hours of debugging: dead transistor, dead LED, dead capacitor, transistors without enough drain current, soldering issues, misunderstandings, transistor current and voltage limitations.

Various Software | [GitHub](#)

2024 - Present

12+ apps shipped: WorkableCafes (cafe wifi and outlet ranker), Pause (native macOS break enforcer), debuff.dev (real-time pair programming), referral.bike (referral tracking), Discord Friends Graph (published on the Chrome Web Store).

Machine Learning From Scratch

2024 - 2025

Quick, Draw! Classification and Generation: Developed CNN classifier reaching 92% accuracy on 95-class dataset (50k sketches per class). Replicated RNN-based Variational Autoencoder in Ha & Eck (2018) for generating human-style vector sketches. ([Demo](#))

DNN for MNIST: Implemented matrix calculus backpropagation to get 95% accuracy and beat a friend's evolutionary algorithm.

Selected Honors

Research Fellow (\$5000 grant) — iph.so

Jul 2026

Building neural representations for video compression to better understand latent physics simulation models.

Finalist (top 5 of 1091 nationally) — IMMC (International Math Modeling Challenge) Nationals

March 2026

Optimized first responder rescue and sweeping paths through various multi-floor buildings with conditions like smoke and fire.

2nd of 16 teams nationally — United States Association for Young Physicists Tournaments (USAYPT)

Feb 2026

Formalized eddy current drag forces with experimental constructions and theoretical calculation.

1st of 29 (\$1500 prize) — TRAE SOLO Hackathon

Oct 2025

Built personal memory layer in 6 hours. Data flows from Omi dev kit recording to graph database to output in search or MCP.

1st of 135 (U1600) — US Amateur Team West (USATW) Chess Tournament

Feb 2024

Scored a perfect score (6/6). Was rated 1270 at the time.

Skills

Learning: Learned algebra i, geometry, algebra ii, and precalculus in 8th grade, learned web, ml, algorithms, simulation in two years, learned piano, frisbee, ukulele, singing in one year no teachers, rubix's cube (pb 21s), colemak, public speaking, more [here](#).

Video Editing and Graphics: Adobe Premiere Pro, Adobe Illustrator, Adobe Photoshop (made 66 videos for fun; [youtube](#))

Teaching: Was a math tutor at Mathnasium for 9 months, taught a friend precalculus in 15 morning sessions, peer tutor at school.

Leadership: Varsity tennis and frisbee team captain, math and chess club lead. Started a discord server that has 2/9 of my school.